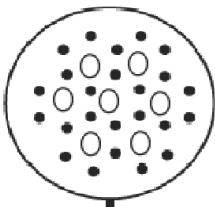


Question				Marking details	Marks Available					
					AO1	AO2	AO3	Total	Maths	Prac
8	(a)	(i)		 Dots showing myosin filaments in correct position and an organised array of actin around the myosin		1		1		
		(ii)		- The actin {moves / slides} over the myosin / actin pulled by myosin (1) - Causing the H band to {shorten / get narrower} (1) - The myosin filaments do not shorten/change length (during contraction) / A band {shows / is} length of myosin filaments (1) - So the A band remains the same length (1)	2	2		4		
		(iii)		57.14 / 57.1 / 57 = 2 marks Award 1 mark for sight of $(3.5-1.5) / 3.5 \times 100$		2		2	2	
		(iv)		Structure of actin: Any one (x1) of: {Actin / it} has myosin <u>binding sites</u> (1) Contraction: Calcium ions bind to troponin causing {troponin / tropomyosin} to {move / change shape} (1) Cross bridges are unable to form at rest: Any one (x1) of: Binding site is {covered / blocked} by tropomyosin when relaxed (1) No calcium ions bind to troponin (1) tropomyosin blocks myosin heads from forming cross bridges. (1)	2			2		